

A general simplified model of symporter with three substrates	
$aA_M + bB_M + cC_M \rightleftharpoons aA_N + bB_N + cC_N$	
Net Transport Flux from M to N	
$J_{symporter}^{M,N(net)} = P_{symporter} \frac{[A]_M^a [B]_M^b [C]_M^c - [A]_N^a [B]_N^b [C]_N^c}{[\alpha^N + 1]^a [\beta^N + 1]^b [\gamma^N + 1]^c}$	(61a)
$J_{A,symporter}^{M,N(net)} = a J_{symporter}^{M,N(net)}$	(61b)
$J_{B,symporter}^{M,N(net)} = b J_{symporter}^{M,N(net)}$	(61c)
$J_{C,symporter}^{M,N(net)} = c J_{symporter}^{M,N}$	(61d)
where $P_{symporter} = [E]_{tg_{symporter}}$ is the membrane permeability to the symporter and its value can be found experimentally [135]. The thermodynamic parameters are defined as: $\alpha^M = \frac{[A]_M}{K_A}, \beta^M = \frac{[B]_M}{K_B}, \gamma^M = \frac{[C]_M}{K_C} \quad \alpha^N = \frac{[A]_N}{K_A}, \beta^N = \frac{[B]_N}{K_B}, \gamma^N = \frac{[C]_N}{K_C}$	

Figure 17: Symporter simplified model. The following assumptions are used: (1) the rapid binding and unbinding assumption, (2) the [quasi-steady-state](#) assumption, (3) symmetric carrier assumption (similarity for all rate constants), (4) symporter has a single site which bound the transported solute, (5) the transported solute competed for the same binding sites, and (6) finally, the empty symporter does not cross the membrane (there is no slippage of empty carrier).

533 5.2.4. Antiporter transport mechanism ($J_{antiporter}$):

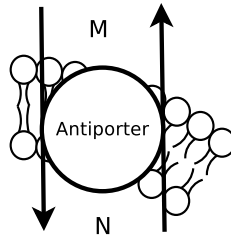


Figure 18: Antiporter Transport Mechanism

534 Antiporters, which are also known as exchangers, are modeled very similarly to symporters. This
535 section covers several commonly used antiporter models, namely, the *eight-state antiporter or-*
536 *dered binding* model, *six-state antiporter ordered binding* model, *competitive antiporter ordered-*